Dataset preparation II:

Data transformation

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Recap

- We have seen that data for *Data mining* should be represented as a **table**.
- We have seen ways to represent non-tabular data into tables: Images, temporal series, documents, transactions, etc.
- We have seen how to inspect your data:
  - Visually: *Boxplots*, *Scatterplots*
  - Numerically: Correlation, regression
- And clean your raw data:
  - Find and impute *Missing Data*
  - Finding and correcting *errors*
  - Finding and considering *outliers*

Today

- Transformation of data:
  - Values in the table
  - Columns of the table
  - Rows of the table

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Value transformation

Value transformation

- It means to change the values in the table
- Several reason to do that:
  - Data homogenization
  - Re-scaling of data
  - Change of distribution

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Data homogenization: Categorical to Numerical

- We have seen that there are a lot of possible values for a column:
  - Numerical
  - Categorical
- Some Data mining algorithms only work with numerical features
- Different ways to transform Categorical Data into numerical:
  - Label Encoding
  - One-Hot encoding
  - Mean encoding

Ordinal (Label) encoding

- When categories have an implicit order
- Examples:
  - *Age*: Child, Young, Adult, Old
  - *Monthly Income*: "0-4k", "4-10k", "10k-20k", ">20k"

Ordinal (Label) encoding

- Example of transformation for *Age* attribute:

	...	*Age*	...
o_1	...	Child	...
o_2	...	Young	...
o_3	...	Adult	...
o_4	...	Child	...
o_5	...	Old	...
...	...	...	...

$\Longrightarrow$

	...	*Age*	...
o_1	...	1	...
o_2	...	2	...
o_3	...	3	...
o_4	...	1	...
o_5	...	4	...
...	...	...	...

- Coding in python and pandas:

```
Age_dict = {'Child':1, 'Young':2, 'Adult':3, 'Old':4}
df['Age'] = df.Age.map(Age_dict)
```

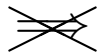
One-Hot encoding

- When categories have no order related to target and there are not a lot of categories (f.i. *color*)

One-Hot encoding

- When categories have no order related to target and there are not a lot of categories (f.i. *color*)
- Ordinal encoding is not right because it induces different distances between modalities

	...	*color*	...
o_1	...	red	...
o_2	...	blue	...
o_3	...	brown	...
o_4	...	red	...
o_5	...	green	...
...	...	...	...



	...	*color*	...
o_1	...	1	...
o_2	...	2	...
o_3	...	3	...
o_4	...	1	...
o_5	...	4	...
...	...	...	...

One-Hot encoding

- One-Hot encoding: Create *dummy* variables, one for modality

	...	*color*	...
o_1	...	red	...
o_2	...	blue	...
o_3	...	brown	...
o_4	...	red	...
o_5	...	green	...
...	...	...	...

$\Longrightarrow$

	...	*red?*	*blue?*	*brown?*	*green?*	...
o_1	...	1	0	0	0	...
o_2	...	0	1	0	0	...
o_3	...	0	0	1	0	...
o_4	...	1	0	0	0	...
o_5	...	0	0	0	1	...
...	...	...	...	...	...	...

- When too many modalities you can group some values to reduce number of modalities (f.i. Grouping categorical values when there is a hierarchical structure over them - *kind of illness* example).
- Always for binary categories (only one column 0-1)
- Coding in `python` and `pandas`:

```
df2 = pd.get_dummies(df, columns =['color'])
```

Mean encoding

- When goal is to predict a label, modalities can be transformed according to joint appearance with labels

	...	*color*	...	*label*
o_1	...	red	...	yes
o_2	...	blue	...	yes
o_3	...	brown	...	no
o_4	...	red	...	no
o_5	...	green	...	yes
...	...	...	...	...

$\Longrightarrow$

	...	*color*	...	*label*
o_1	...	0.5	...	yes
o_2	...	1	...	yes
o_3	...	0	...	no
o_4	...	0.5	...	no
o_5	...	1	...	yes
...	...	...	...	...

Categorical to numeric

- Not the whole story. A lot of other methods depending on the data.

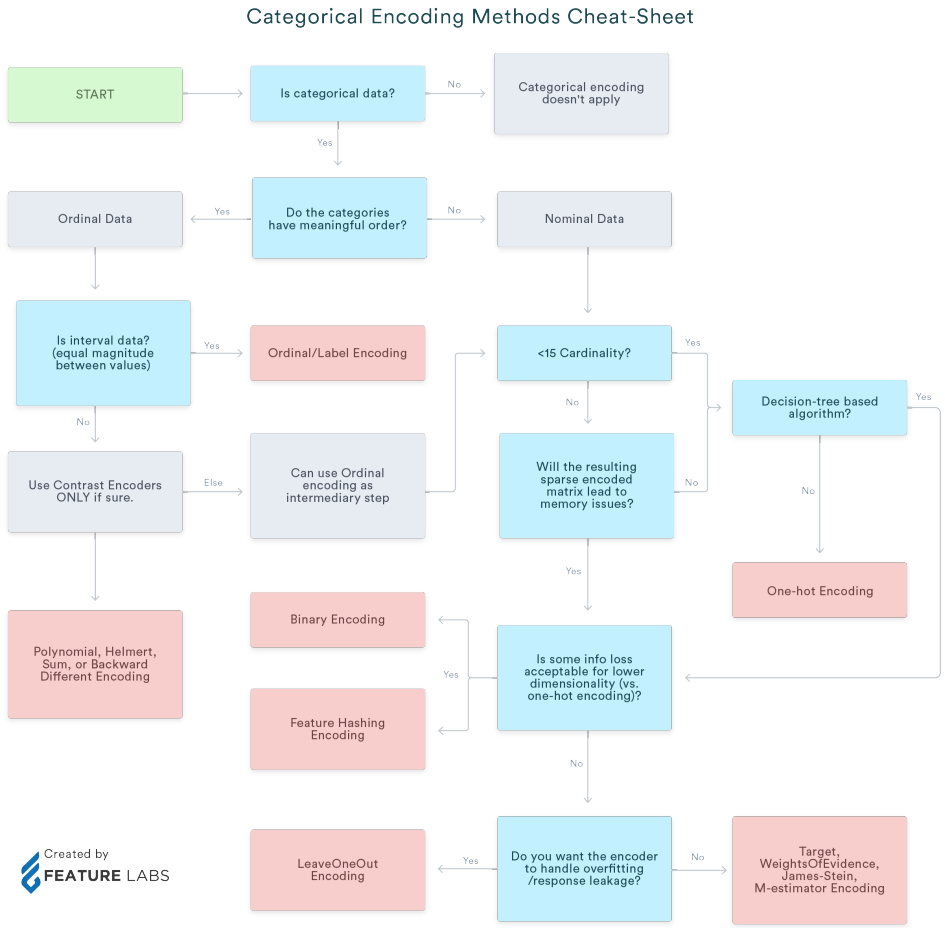


Figure: source

Data homogenization: Numerical to Qualitative

- From qualitative to numerical
- Reducing number of values:
  - From numerical to ranges (f.i. age: 0-18, 19-40, 40-65, 67-120)
  - From numerical to qualitative (f.i. age: young, adult, senior, old)

Value transformation

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Transforming values of columns

- All variables should be in the same range to avoid bias in computation of distances.
- Mandatory for some algorithms like *k-NN*, *SVM* or Neural Networks
- Can be seen as change of units
- Common procedures: Normalization or Standardization
- Caveat: Different assumptions!

Transforming values of columns

- Normalization $[0 - 1]$:

$$v(i) = \frac{v(i) - min(v)}{max(v) - min(v)}$$

```
from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler()
scaler.fit(data)
d2 = scaler.transform(data)
```

- Standardization $N(0, 1)$:

$$v(i) = \frac{v(i) - mean(v)}{std(v)}$$

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
scaler.fit(data)
d2 = scaler.transform(data)
```

Value transformation

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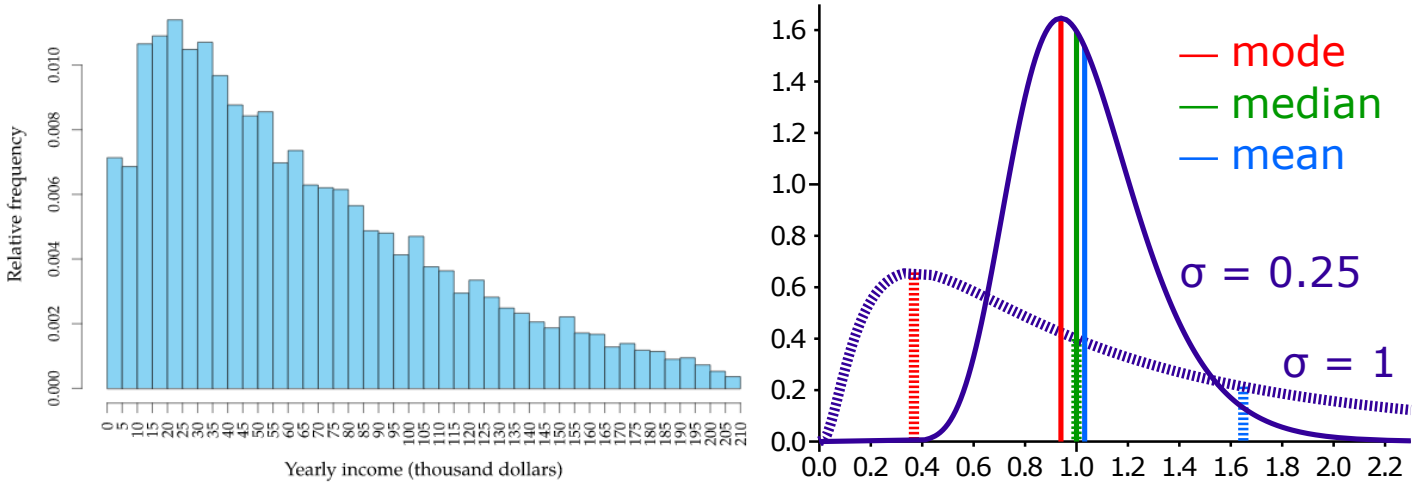
Value transformation

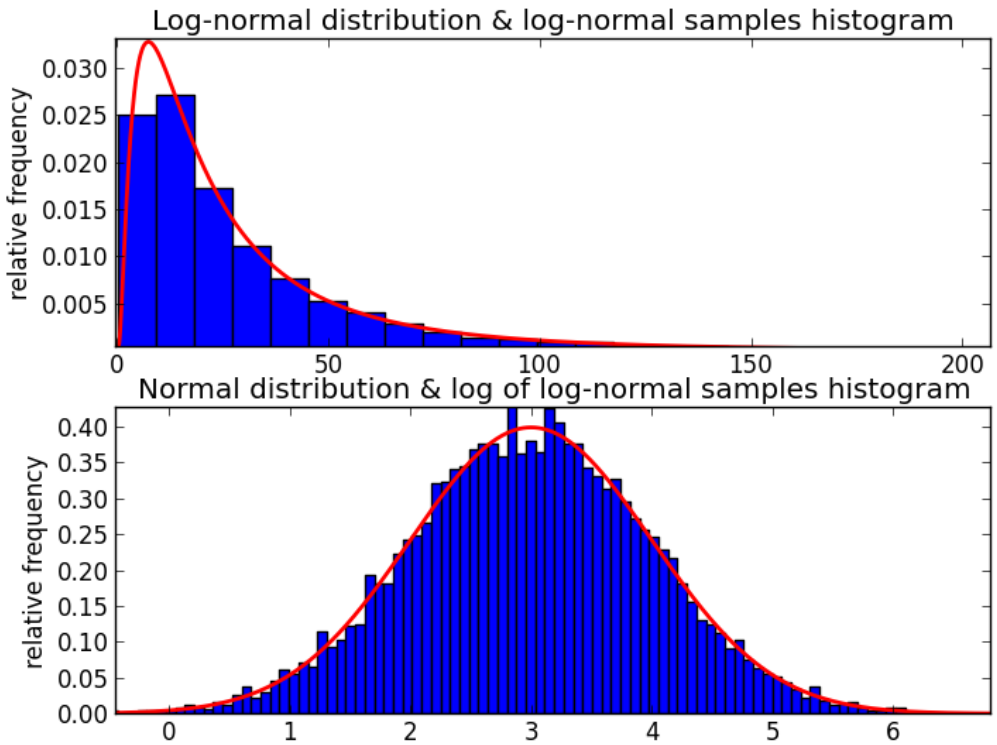
- In some cases distribution of data has heavy tails



- In these cases, we have problems distinguishing between most of data.
- Data appear as *outliers*
- Solutions:
  - Apply Log of column (log-normalization)
  - Replace values by quantiles

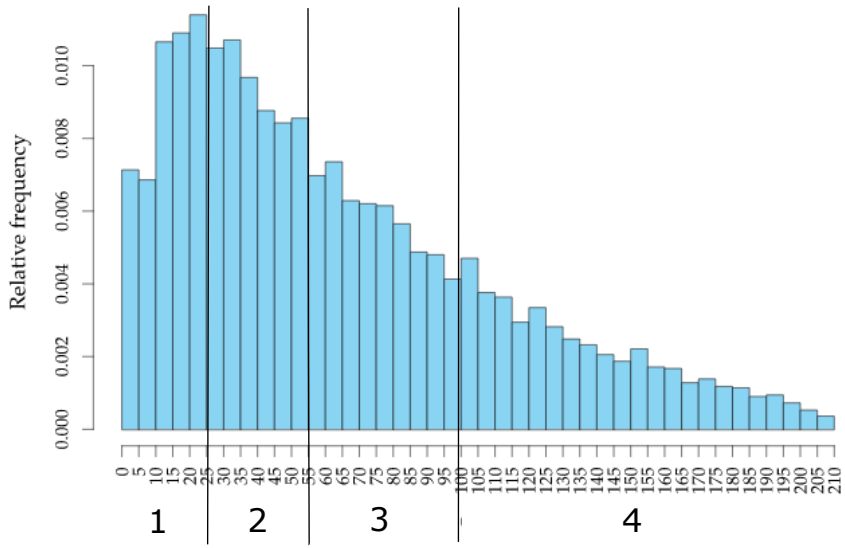
Value transformation

- Effect of log-normalization:



Value transformation

- Replace value according the position in the quantile
- Effect of quantiles separation when using 4 values



- You can use a higher degree of quantization if needed

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Work on columns

Subsection 1

Reducing number of columns: feature selection and feature extraction

Removing variables

- Very common in DM
- Why dimensionality reduction?
  1. Reduced Computing Time
  2. Increased predictive/descriptive accuracy
  3. Better representation of the data-mining model
  4. The intrinsic dimension may be small.
- Some data mining techniques may not be effective for high-dimensional data (they suffer the *Curse of dimensionality*: accuracy and efficiency may degrade rapidly as the dimension) increases.

Reducing number of columns

Two ways to reduce the number of columns:

1. **Feature selection**: A process that chooses an optimal subset of features according to an objective function
   - Feature ranking algorithms, and
   - Minimum subset algorithms.
2. **Feature extraction**: refers to the mapping of the original high-dimensional data onto a lower-dimensional space.
   - Descriptive setting: minimize the information loss
   - Predictive setting: maximize the class discrimination

Removing variables: Feature selection

Feature selection: Filter and feature ranking algorithms.

- Remove columns without enough variability ($std(v) < threshold$)
- Remove variables that are not enough correlated with label ($corr(v, label) < threshold$)
- Remove variables with low *Fisher score* with label ($F(v, label) < threshold$)
- Remove variables without enough Information Gain ($IG(v) < threshold$)
- Remove variables without enough Relief [Paper] ($Relief(v) < threshold$)

All these methods implicitly build a ranking of features using a given measure. After ranking, we use a threshold to select the interesting features.

Removing variables: Ranking or filter methods

Let X_v^l the set of values of feature v with label l and $n_{v,l}$ the number of values of that set.

- *Fisher Score* computes differences in values for different labels:

$$F_v = \frac{|\ (\mu(X_v^1) - \mu(X_v^0)\ |}{\sqrt{\frac{\sigma(X_v^1)}{n_{v,1}} + \frac{\sigma(X_v^0)}{n_{v,0}}}}$$

- Example:

v_1	v_2	l
0.3	0.7	0
0.2	0.9	1
0.6	0.6	0
0.5	0.5	0
0.7	0.7	1
0.4	0.9	1

$$X_{v_1}^0 = \{0.3, 0.6, 0.5\}$$
$$X_{v_1}^1 = \{0.2, 0.7, 0.4\}$$
$$X_{v_2}^0 = \{0.7, 0.6, 0.5\}$$
$$X_{v_2}^1 = \{0.9, 0.7, 0.9\}$$

Removing variables: Ranking or filter methods

$$\sqrt{\frac{\sigma(X_{v_1}^1)}{n_{v_1,1}} + \frac{\sigma(X_{v_1}^0)}{n_{v_1,0}}} = \sqrt{\frac{0.0233}{3} + \frac{0.6333}{3}} = 0.4678$$

$$\sqrt{\frac{\sigma(X_{v_2}^1)}{n_{v_2,1}} + \frac{\sigma(X_{v_2}^0)}{n_{v_2,0}}} = \sqrt{\frac{0.01}{3} + \frac{0.0133}{3}} = 0.0875$$

- Test with threshold 0.5:

$$\frac{|\,(\mu(X_{v_1}^1) - \mu(X_{v_1}^0)\,|}{\sqrt{\frac{\sigma(X_{v_1}^1)}{n_{v_1,1}} + \frac{\sigma(X_{v_1}^0)}{n_{v_1,0}}}} = \frac{|\,0.4333 - 0.4667\,|}{0.4678} = 0.0735 < 0.5$$

$$\frac{|\,(\mu(X_{v_2}^1) - \mu(X_{v_2}^0)\,|}{\sqrt{\frac{\sigma(X_{v_2}^1)}{n_{v_2,1}} + \frac{\sigma(X_{v_2}^0)}{n_{v_2,0}}}} = \frac{|\,0.8333–0.6\,|}{0.0875} = 2.6667 > 0.5$$

Removing variables: Ranking or filter methods

- Correlation:

$$\rho(v, I) = \frac{cov(v, I)}{\sigma_v \sigma_I}$$

- Information Gain (Mutual Information): when v and I are independent variables $p(v, I) = p(v)p(I)$

$$I(v; I) = \sum_{x \in \mathcal{D}} p(x_v, x_I) \log \frac{p(x_v, x_I)}{p(x_v)p(x_I)} = H(I) - H(I|v)$$

where H is entropy measure

Removing variables: Ranking or filter methods

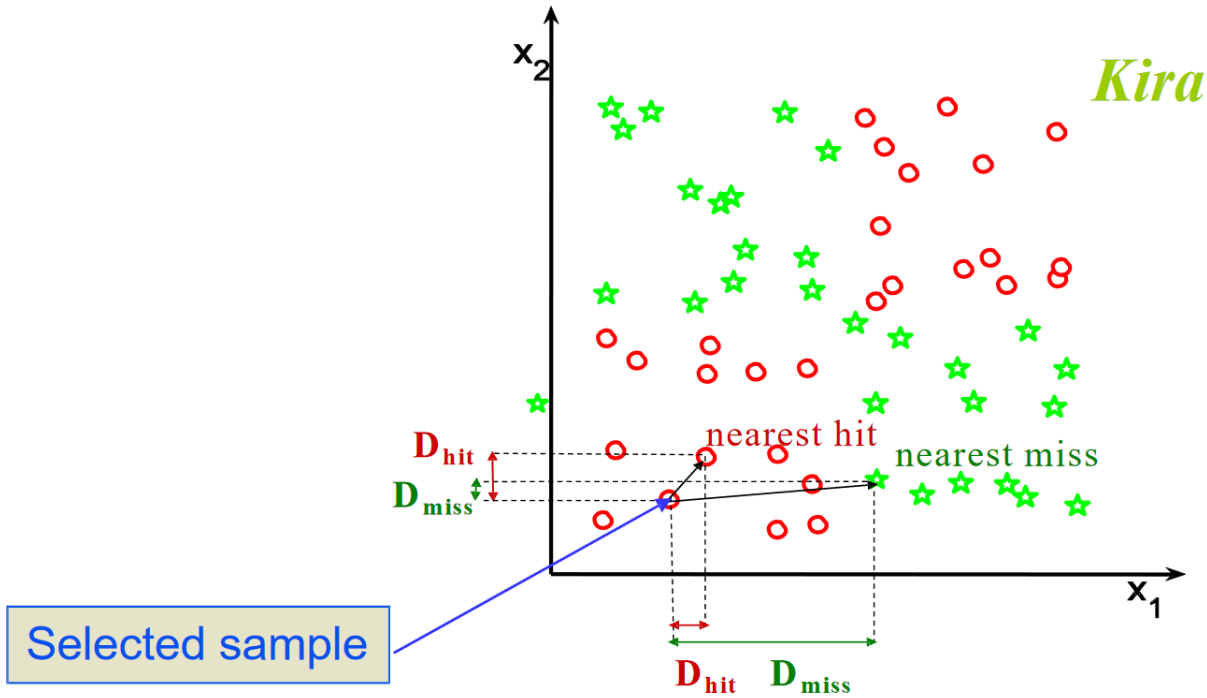
Relief algorithm:

- Basic algorithm construct:
  - Each feature is assigned with cumulative weight computed over a subset of the training data set.
  - Feature with weight over a certain threshold is the selected feature subset.
- Weight computation:
  - Let X be a randomly selected sample
  - *near-hit* sample = the closest sample from the same class
  - *near-miss* sample = the closest sample from the different class.
  - Impact on weight for feature j of example X is computed as:

$$W_j = -diff_j(X, near\text{-}hit)^2 + diff_j(X, near\text{-}miss)^2$$

  - Procedure repeated for several random samples

Removing variables: Ranking or filter methods



Kira and Rendell, 1992

Algorithm 1 : Pseudocode for **Relief** algorithm.

Input:

N number of features

m number of instances sampled

τ adjustable threshold

initialize w=0

for $i = 1$ to m **do**

$I \leftarrow$ randomly selected instance of X

Find nearest-hit H and nearest-miss J of I

for $j = 1$ to N **do**

$w(j) \leftarrow w(j) - diff_j(X, near\text{-}hit)^2/m + diff_j(X, near\text{-}miss)^2/m$

end for

end for

Output: features j with $w(j) > \tau$

Removing variables: Redundant features

- We have seen how to reduce features irrelevant to the learning task

Removing variables: Redundant features

- We have seen how to reduce features irrelevant to the learning task
- We can also reduce the number of features removing **redundant** features
- Redundant features are those that show similar values for each instance (row) *after* normalization
- You can use some of the proposed measures (f.i. correlation, MI) and thresholds to detect redundant features. When two features are redundant, remove one of them.

Removing variables: Redundant features

- We have seen how to reduce features irrelevant to the learning task
- We can also reduce the number of features removing **redundant** features
- Redundant features are those that show similar values for each instance (row) *after* normalization
- You can use some of the proposed measures (f.i. correlation, MI) and thresholds to detect redundant features. When two features are redundant, remove one of them.
- However, it is time consuming because you have tot test each pair of features (cost $O(f^2)$).
- In general redundant features do not hurt the performance of models, only impacts in running time and interpretation of models.

Removing variables: Feature selection

- How to select threshold?

Removing variables: Feature selection

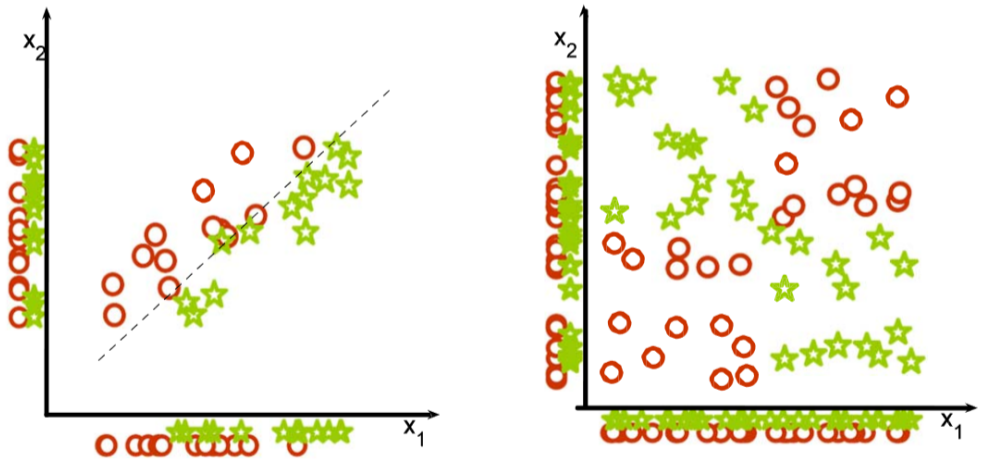
- How to select threshold? (1) Rank features and use top n in ranking

Removing variables: Feature selection

- How to select threshold? (1) Rank features and use top n in ranking or (2) plot features against values and choose threshold when values decrease slowly

Removing variables: Feature selection

- How to select threshold? (1) Rank features and use top n in ranking or (2) plot features against values and choose threshold when values decrease slowly
- Rank and cut methods could miss joint effects of variables:

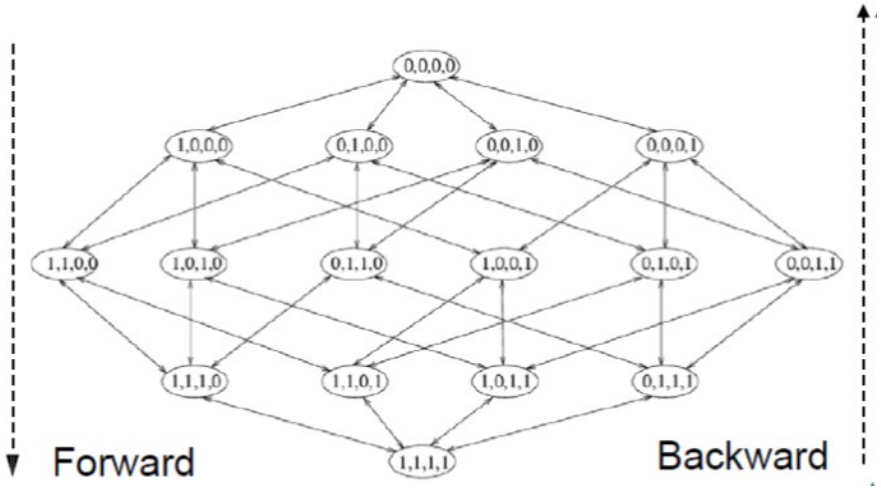


Guyon-Elisseeff, JMLR 2004; Springer 2006

Removing variables: Feature selection

Instead of evaluating features isolated, we will evaluate sets of features:

- It can be viewed as a search problem (2^n sets of features to consider)



- An optimal search is not feasible, and simplifications are made to produce acceptable and timely reasonable results:
  - heuristic criteria
  - bottom-up approach
  - top-down approach

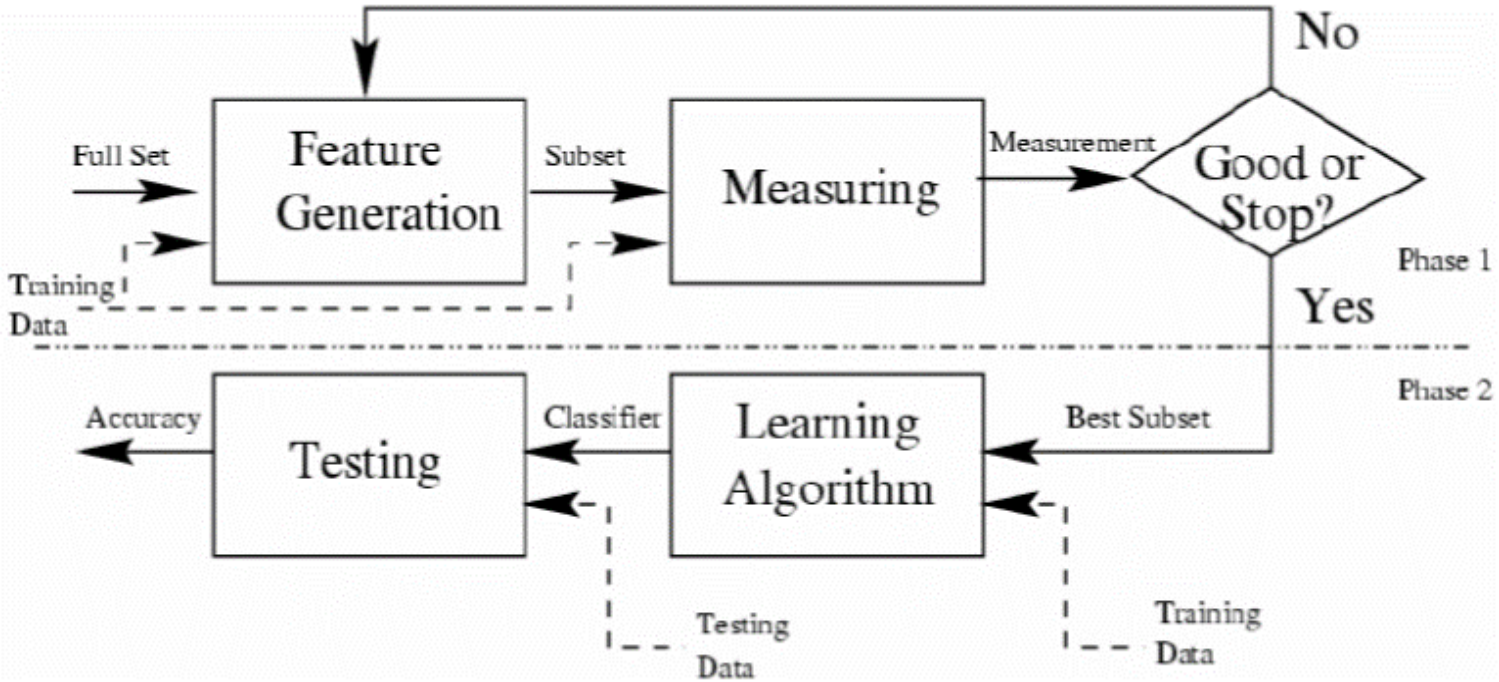
Removing variables: Feature selection

Advanced feature selection methods have the following elements:

- A Feature set generator that proposes a set of features to evaluate
- An evaluation method. Two kinds of evaluation methods:
  1. *Filter*: Uses an evaluation metric that is independent of the learning mechanism that will be used (f.i. the ones we have seen in the previous slides)
  2. *Wrapper* method: Uses a measure of the target learning method to evaluate the set of features (f.i. in SVM, the margin of the classifier obtained with that set of features)
- A criteria to stop the search for the set of features

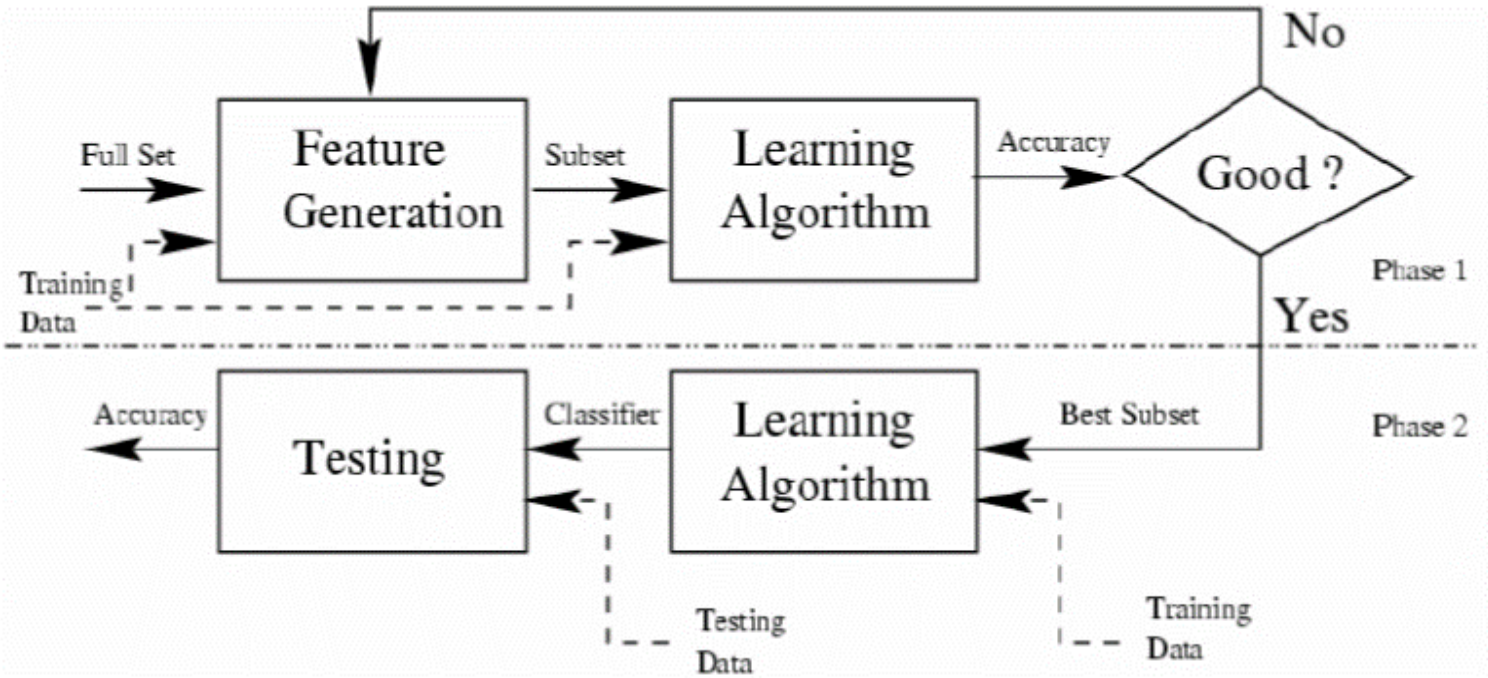
Removing variables: Feature selection

Filter method:



Removing variables: Feature selection

Wrapper method:



Reducing number of columns

Two ways to reduce the number of columns:

1. **Feature selection**: A process that chooses an optimal subset of features according to an objective function
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Reducing number of columns: Feature extraction

- **Feature extraction**: refers to the mapping of the original high-dimensional data onto a *lower-dimensional space*.
- The new dimensions (columns) are a *new* and should contain the information of the original dataset in a lower number of columns.
- Examples:
  - **PCA** (descriptive setting): minimize the information loss embedding the dataset in a space of new columns that are build as linear combinations of the original columns. Columns are sorted by inertia, and we keep only the n most informative columns.
  - **LDA** (predictive setting): does like in the case of PCA an embedding in a dimensional space of columns that are build as linear combinations of the original columns, but in this case the goal is to maximize the class discrimination instead of minimal information loss.

Subsection 2

Enriching the data set: Joining tables

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- It is common to have information stored in data bases
- Databases, for instance relational methods, store information in separate tables.
- So you can complete (enrich) your dataset with corresponding data from another table

Enriching the data set: Joining tables

- It is common to have information stored in data bases
- Databases, for instance relational methods, store information in separate tables.
- So you can complete (enrich) your dataset with corresponding data from another table
- For instance think in a dataset about pulmonary diseases. It has several columns plus the identifier of the patient.
- In other dataset you have information about each patient. For instance residence
- Joining both tables you now can relate features of the first dataset with residence, for instance.

Subsection 3

Enriching the data set: Building new variables

Building new variables: Ratios and Proportions

- Sometimes your data is expressed in columns that it is difficult to manipulate with your data mining algorithm. So you add define new columns from the old ones
- For instance think in a dataset about taxis, with columns like departure point, destination point, duration of trip.
- You can add to to the data set new columns: length of trip, speed of taxi
- So you can complete (enrich) your dataset with columns that ease the finding of a good model

Building new variables: Automated creation

- Some algorithms only work with lineal dependencies of variables.
- Generate a new feature set consisting of all "polynomial" combinations of the features with degree less than or equal to the specified degree.
- For example, from three initial features

$$\{X_1, X_2, X_3\}$$

add set of new variables

$$\{X_1X_2, X_1X_3, X_2X_3, X_1X_2X_3\}$$

with a degree of 3.

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Data processing: rows

Subsection 1

Reducing the number of rows

Reducing the number of rows

- Most DM algorithms have time complexity cost $O(n^2)$ (f.i. kNN) or even $O(n^3)$ (f.i. SVM), where n is the number of rows in the dataset.
- Some algorithms need to store matrices of distances between elements, so we need $O(n^2)$ space.
- So, in some datasets we could have too many rows to build a model in reasonable time
- ... but maybe not all rows are necessary to build a model.
- So we could try to reduce the number of elements using a *sampling* procedure

Reducing the number of rows

Some questions appear when considering sampling:

- How many examples should I sample?
- Which kind of sampling should I use
- Am I sure that the sampling was successful?

Sampling procedure

Which kind of sampling should I use? Different kinds of sampling

- *Systematic sampling*: get every k-th sample in the dataset. For instance if I want 50% sampling, get one and the next one no.
- **Problem!**: when there are regularities in data set (data set sorted, for instance)

Sampling procedure

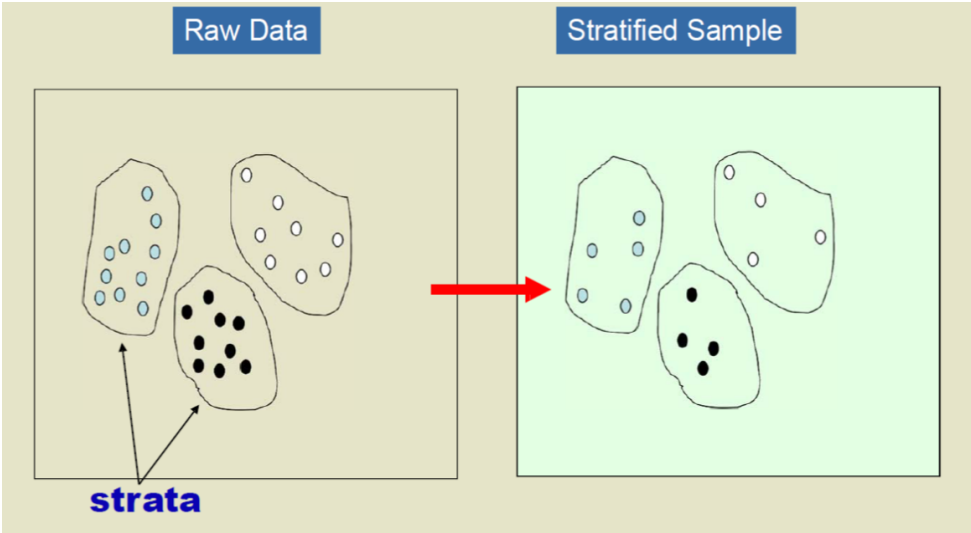
Which kind of sampling should I use? Different kinds of sampling

- *Systematic sampling*: get every k-th sample in the dataset. For instance if I want 50% sampling, get one and the next one no.
- **Problem!**: when there are regularities in data set (data set sorted, for instance)
- *Random sampling*: Two different choices
  - random sampling without replacement,
  - random sampling with replacement.
- **Problem!**: Is my sampling representative of the complete dataset?

Sampling procedure

Which kind of sampling should I use? Different kinds of sampling

- *Stratified sampling*: Split data set into non-overlapping subsets (strata) and combine strata results.
- Used in general to ensure same proportion of each class of examples in the sample.



- **Problem!**: Ok. Class values are representative, but What happens with other columns?

Sampling procedure

Which kind of sampling should I use? Different kinds of sampling

- *Multi-Stratified sampling*: Do a Stratified sampling, and after that, *verify* that for each column we have a similar statistical structure to the original dataset (histogram of modalities in qualitative features, mean for numerical values). If the sampled set does not pass the test, throw it and do another sampling until one that passes the test is found.

Number of examples

How many examples should I sample?

8000 points 2000 Points 500 Points

- Error should be not different than using complete dataset.
- One solution could be **incremental sampling**: In consists in starting with a small sample, measure the accuracy with the learning algorithm and increase the size of the sampling until no improvement in accuracy is found.

Number of examples

How many examples should I sample?

- Data should be representative of the original dataset.
- **Inverse sampling**: Used when features with rare frequencies of values (skewed distributions): Consists in the dynamic increase of examples sampled until some condition about feature are satisfied (f.i. all values appear at least t times)

Number of examples

- With sampling procedures we usually waste examples. One approach that could help in polynomial cost algorithms is to use ensemble techniques.
- Assume a SVM with training cost $O(n^3)$ and $n = 10^6$ examples:

$$O((10^6)^3) = 10^{18}$$

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- Consider the following procedure:
  - Divide the dataset in 10 parts
  - Build an SVMs on each part (we have 10 SVMs)
  - Combine by *majority vote* the output of each classifier in testing

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- Consider the following procedure:
  - Divide the dataset in 10 parts
  - Build an SVMs on each part (we have 10 SVMs)
  - Combine by *majority vote* the output of each classifier in testing
  - Now training cost is:

$$\sum_{i=1}^{10} O((10^5)^3) = 10^{16}$$

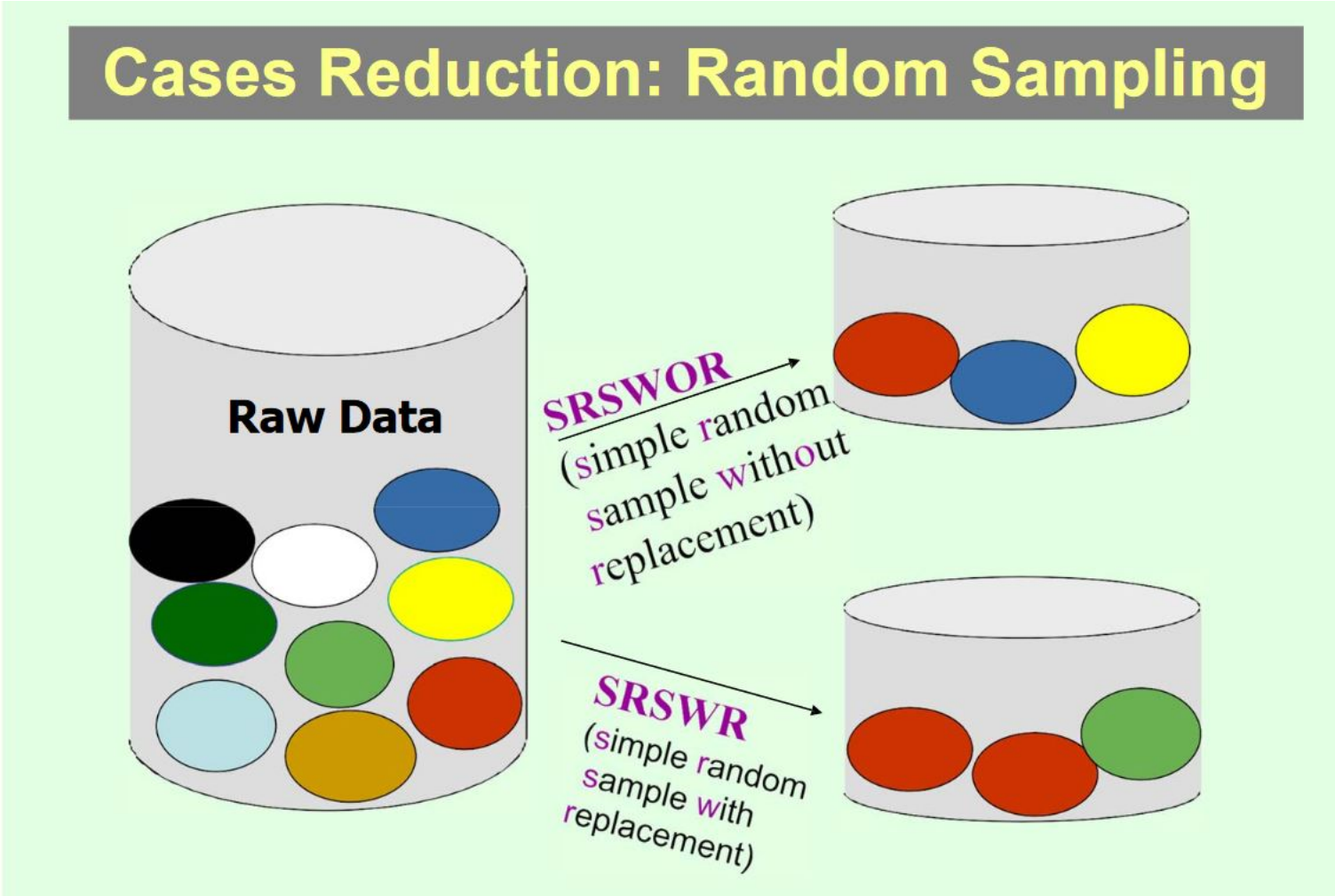
Subsection 2

Increasing the number of rows

Number of examples

- In some cases we don't have enough data
  - Small datasets
  - Unbalanced datasets: We have several examples but one of the classes has very few examples
- We need more data, but how to generate data?
- Two strategies: resampling and data generation

Sampling procedures



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- Try to use a cost function when building the classifier to penalize different kinds of errors in the confusion matrix

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- Remember always to test on untouched validation dataset.